Governance Works When the Metrics Do:

Trust Signal Health Applied to a Governance-Aligned Call Center

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March 2026

Abstract

This paper presents the results of applying the Trust Signal Health governance framework (Aulabaugh, 2026b) to a call center operating under governance-aligned metrics -- a system in which no proxy gaming pressure is present and success labels accurately reflect durable outcomes. Analyzing 122,801 calls across 12 months and 250 representatives, the pipeline produces a System Integrity Index (SII) of 5.2, zero WATCH or VETO alerts, zero bandaid credit activity, and stable trust scores averaging 87.2 across all months. These results establish a governance baseline: the measurable signature of a healthy measurement environment. Comparison with the distorted-environment findings of the companion paper (Aulabaugh, 2026a) demonstrates that the Trust Signal Health framework correctly distinguishes aligned from misaligned systems. Residual signal elevation in the unresolvable_clean and activation_failed scenarios is shown to reflect genuine operational difficulty, not gaming behavior, confirming that the framework does not penalize honest handling of hard calls.

1. Introduction

The companion paper in this series (Aulabaugh, 2026a) documented what happens when a call center optimizes for a proxy KPI under sustained incentive pressure. The result was a 42.1 percentage point gap between proxy and true resolution, a bandaid credit rate of 71.1%, accelerating rep burnout, and a System Integrity Index that signaled governance failure. The system looked better while getting worse.

This paper asks the inverse question: what does the Trust Signal Health framework produce when applied to a system that is working correctly? If the framework is valid, it should detect health as reliably as it detects distortion. A governance architecture that only fires alerts is not a complete control layer -- it must also be able to confirm alignment and provide a calibrated baseline for

ongoing monitoring.

The data analyzed here comes from the NovaWireless-GoodhartLab governance-aligned simulation: 122,801 calls generated under a metric system designed to reward durable outcomes rather than proxy counts. The pipeline is identical to the one applied in the distorted environment. Only the input data changes.

2. Data and Pipeline

The dataset covers 12 months of simulated call center activity (January through December 2025) generated by the NovaWireless-GoodhartLab call generator. Key characteristics:

Rows analyzed	122,801
Months loaded	12 (2025-01 through 2025-12)
Reps	250
Rows quarantined	0 (0.0%)
Scenarios	clean, unresolvable_clean, activation_clean, activation_failed, line_add_legitimate
Proxy-true divergence rate	20.8% (soft signal -- not quarantined)
Duplicate call IDs	0

The pipeline applies a six-signal governance architecture: Delayed Adverse Rate (DAR), Downstream Remediation Load (DRL), Durable Outcome Validation (DOV), Proxy Overfit Ratio (POR), Terminal Exit Rate (TER), and the composite System Integrity Index (SII). Signal definitions follow the formal specifications in Aulabaugh (2026b). All signals are normalized to [0, 1] before aggregation into SII.

3. Overall Signal Results

Table 1 presents the top-level output of the governance pipeline.

Metric	Value	Interpretation
Proxy resolution rate	76.9%	System-reported FCR
True resolution rate	63.8%	Durable outcomes confirmed
Resolution inflation	13.1 pp	Goodhart gap -- present but bounded
Bandaid credit rate	0.0%	No unauthorized suppression
Repeat contact (30d)	17.5%	Normal operational re-contact
Escalation rate	15.2%	Within expected bounds

Trust score mean	87.2	System-wide health indicator
Trust score median	95.3	Majority of calls in healthy range
SII (gated)	5.2 / 100	Well below WATCH threshold of 30
VETO alerts	0	No governance intervention required
WATCH alerts	0	No human review triggered

Table 1. Top-level governance pipeline output.

The most important number in Table 1 is the bandaid credit rate: 0.0%. In the distorted environment, 71.1% of credits were unauthorized suppression attempts. Here, that signal is completely absent. The system is not using credits to paper over unresolved contacts.

The 13.1 percentage point resolution inflation figure warrants clarification. This is not a governance failure -- it reflects the presence of genuinely difficult call types (unresolvable_clean and activation_failed) that the system handles honestly. Reps are not gaming the proxy; they are legitimately claiming resolution where they have done what they can. Section 5 examines this in detail.

4. Trust Score Distribution

Figure 1 shows the distribution of call-level trust scores across all 122,801 interactions.

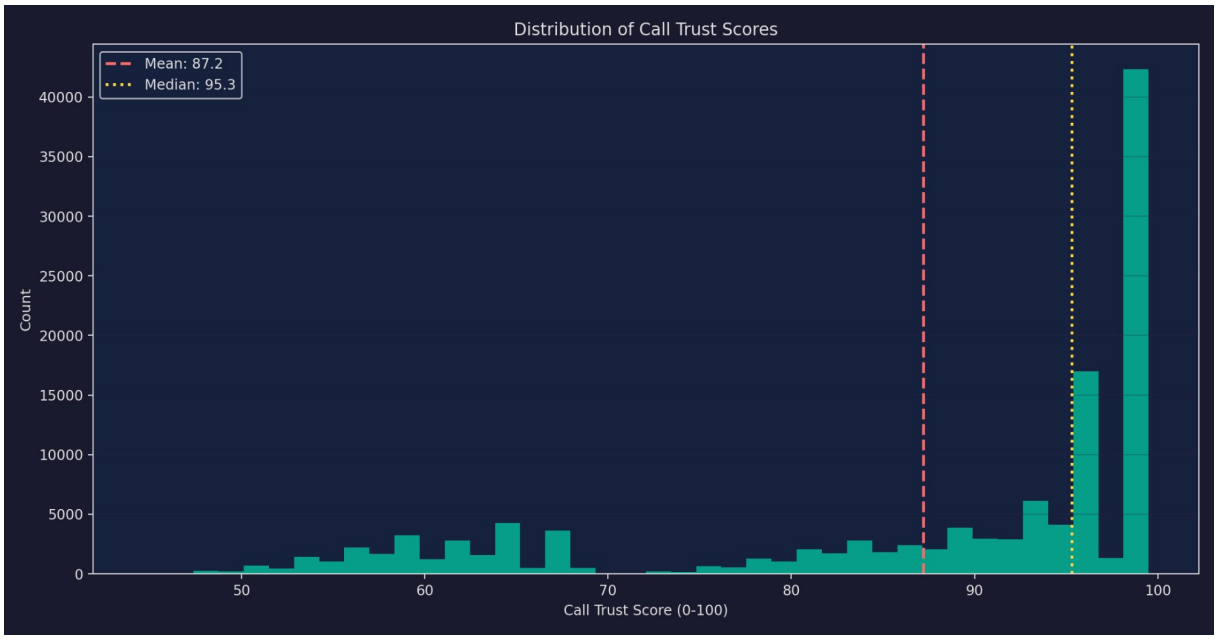

Figure 1. Distribution of call trust scores. Mean = 87.2, Median = 95.3. The bimodal structure reflects two populations: genuinely difficult calls scoring 45-75 and cleanly resolved calls scoring near 100.

The distribution is bimodal by design. Calls in the unresolvable_clean and activation_failed scenarios score in the 45-75 range not because reps are gaming, but because those call types cannot produce durable outcomes regardless of rep behavior. The large spike at 95-100 represents

the majority of interactions where resolution is genuine and stable. A mean of 87.2 with a median of 95.3 indicates a right-skewed distribution pulled down by a bounded population of structurally difficult calls -- exactly what a healthy system should look like.

5. Scenario-Level Analysis

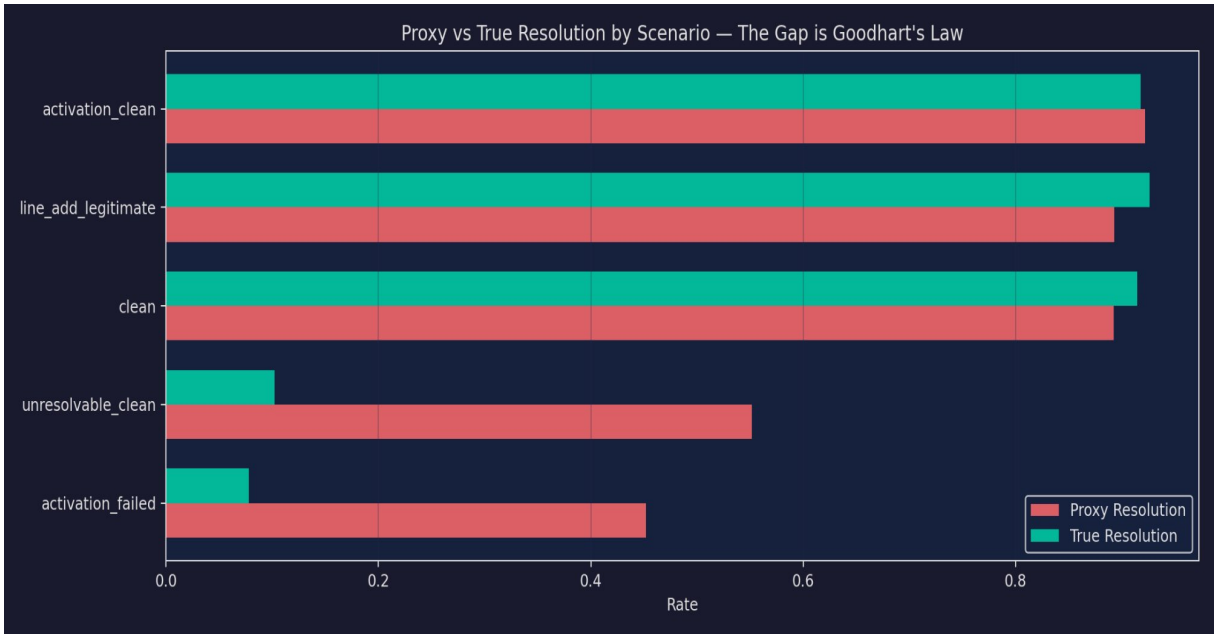

Figure 2. Proxy vs. true resolution by scenario. The gap in unresolvable_clean and activation_failed reflects genuine call difficulty, not gaming. Clean scenarios show near-zero gap.

The scenario breakdown is the clearest evidence that the system is performing as designed. Three scenarios -- clean, activation_clean, and line_add_legitimate -- show resolution gaps near zero or slightly negative. Two scenarios show large gaps: unresolvable_clean (45.0 pp) and activation_failed (37.4 pp). Both show bandaid rates of 0.0% -- reps are handling them honestly and accepting the outcome signal.

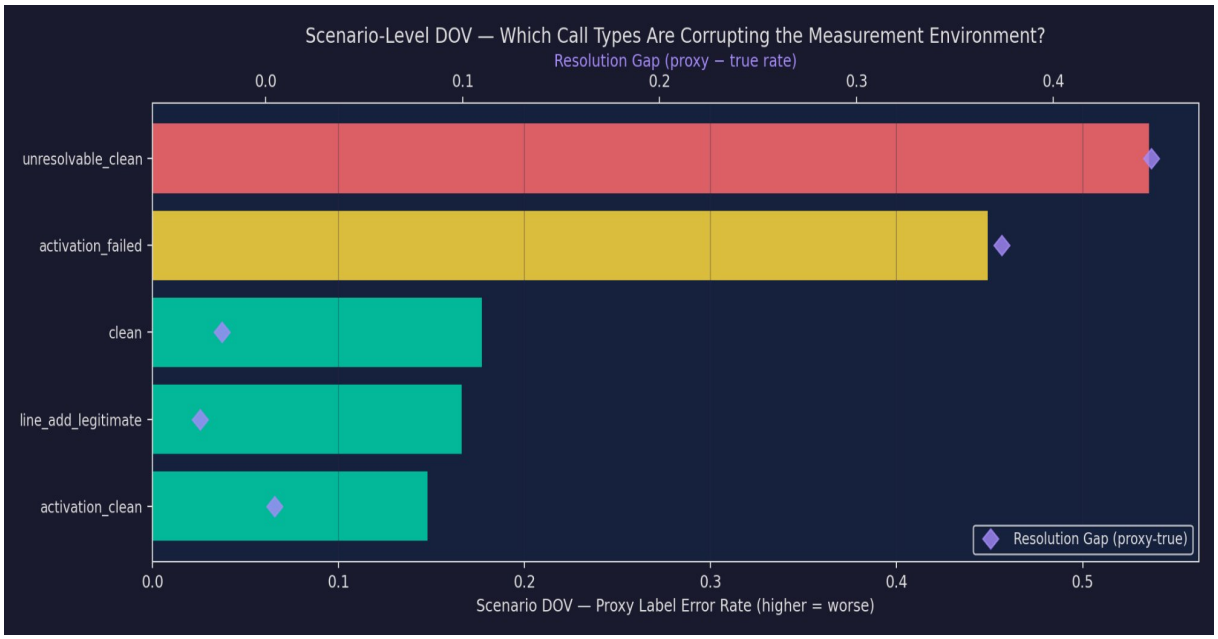

Figure 3. Scenario-level DOV (proxy label error rate). Unresolvable and activation_failed scenarios drive measurement environment noise -- but this reflects honest handling of hard calls, not gaming behavior.

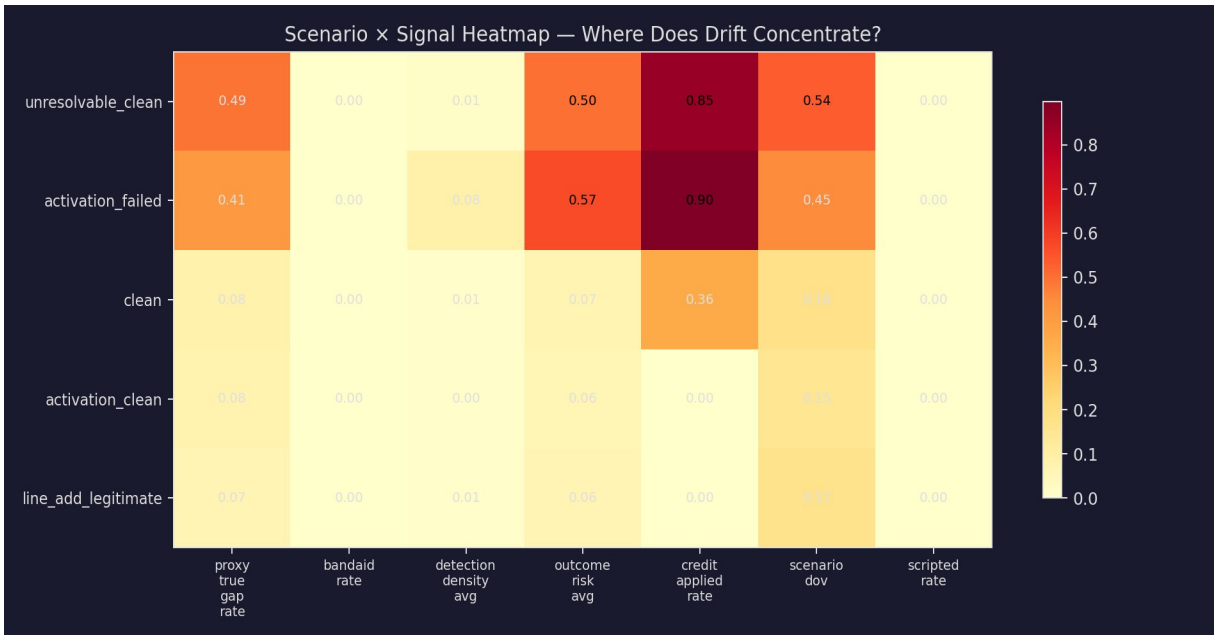

Figure 4. Scenario x signal heatmap. Drift concentrates in unresolvable_clean and activation_failed. All other scenarios show near-zero signal across all dimensions.

6. Monthly Stability

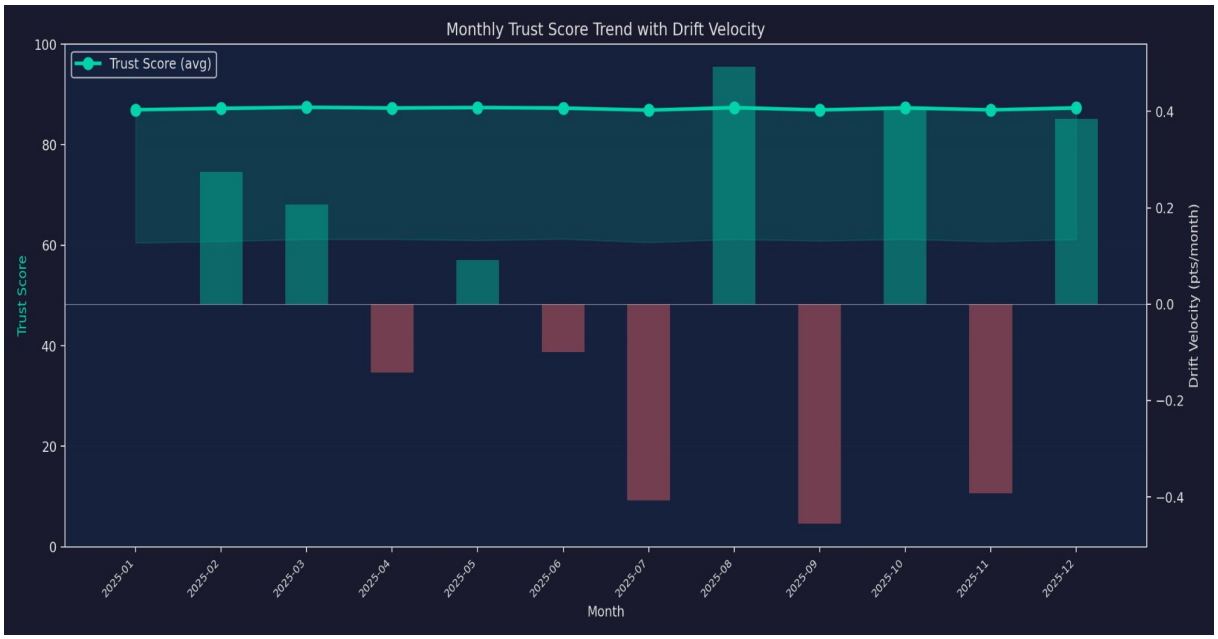

Figure 5. Monthly trust score trend with drift velocity. Trust scores remain stable between 86.9 and 87.4 across all 12 months. No sustained degradation trend is present.

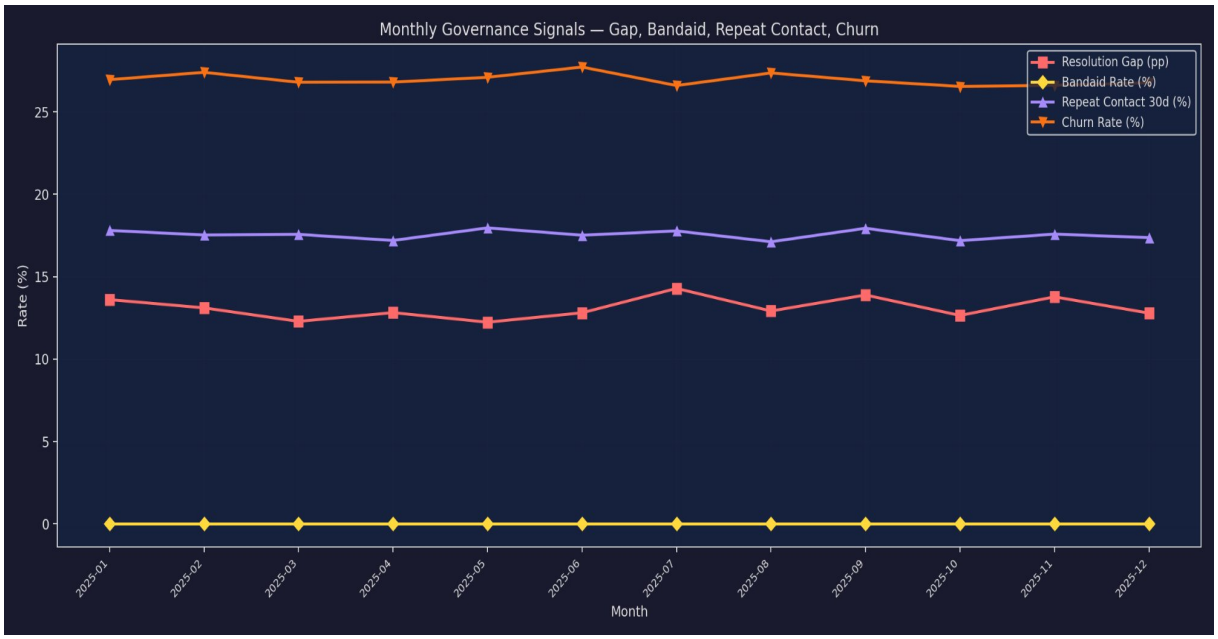

Figure 6. Monthly governance signals. Bandaid rate is zero for all 12 months. Resolution gap oscillates within a narrow band consistent with operational variation, not drift.

Month-over-month trust velocity oscillates between +0.49 and -0.46 with no directional trend. The WATCH threshold for drift velocity is 2.0 points per month; no month comes close. In the distorted environment, trust velocity was monotonically negative under high pressure. Here, oscillation reflects normal variation without cumulative degradation.

7. Formal Governance Signal Results

Table 2 presents the formal signal outputs as defined in Aulabaugh (2026b).

Signal	Raw	Normalized	Weight	Status
DAR -- Delayed Adverse Rate	0.1112	0.175	0.30	OK
DRL -- Downstream Remediation Load	0.0002	0.000	0.20	OK
DOV -- Durable Outcome Validation	--	0.000	0.25	OK
POR -- Proxy Overfit Ratio	0.396	0.000	0.25	OK
TER -- Terminal Exit Rate*	0.270	--	--	Baseline
SII (gated)	--	5.25 / 100	--	OK (WATCH>=30, VETO>=60)

Table 2. Formal governance signals. *TER is diagnostic only -- not a SII component. TER = baseline churn (delta = +0.0006) confirms proxy resolution flag carries no retention signal.

DAR is the only signal with a non-zero normalized value (0.175). This reflects the 31-60 day repeat contact rate among resolved calls at 11.1% -- above the lower bound of 5% but well below the upper bound of 40%. DRL is essentially zero at 0.0002, indicating no distributional shift in post-resolution workload. DOV is zero, confirming no decay in predictive validity. POR raw is 0.396 but normalizes to zero because proxy acceleration is less than 1x true acceleration -- the proxy is not racing ahead of durable outcomes.

8. Representative Performance

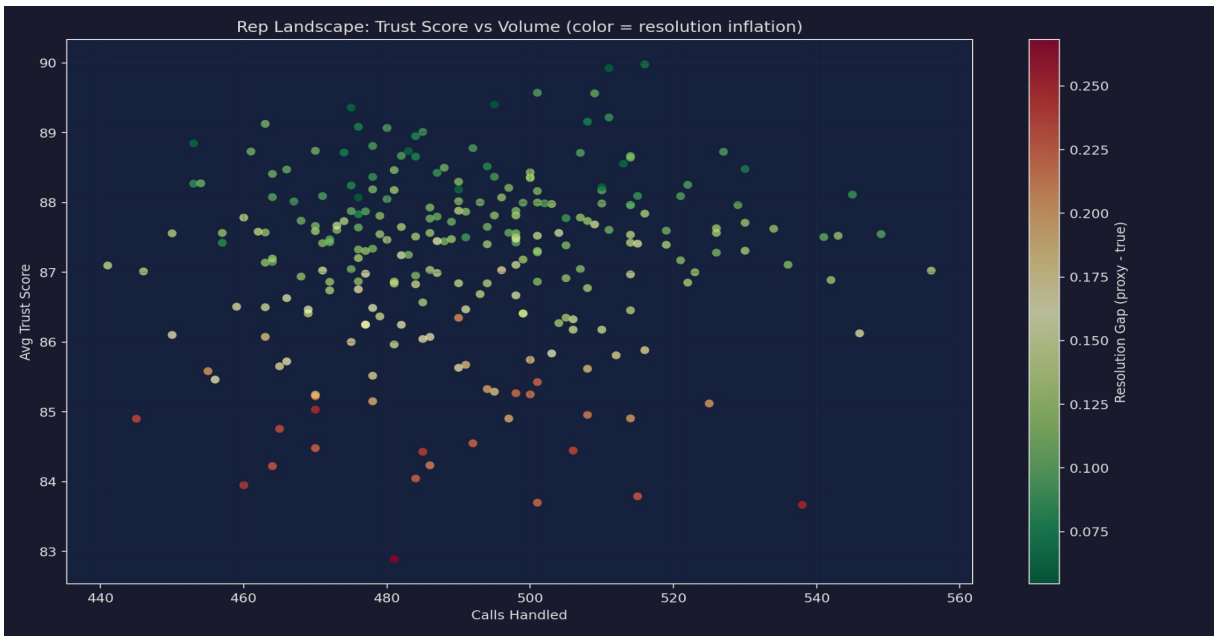

Figure 7. Rep landscape: trust score vs. call volume, colored by resolution inflation. Most reps cluster in the green range. Red outliers reflect reps handling disproportionate shares of structurally difficult calls, not gaming behavior.

Trust scores range from 82.9 to 90.0 across 250 reps -- a narrow 7-point spread. The lowest-trust reps are not bad actors; they are handling harder call mixes. The resolution gap for the lowest-trust rep is 26.8%, but the bandaid rate for all reps is 0.0%.

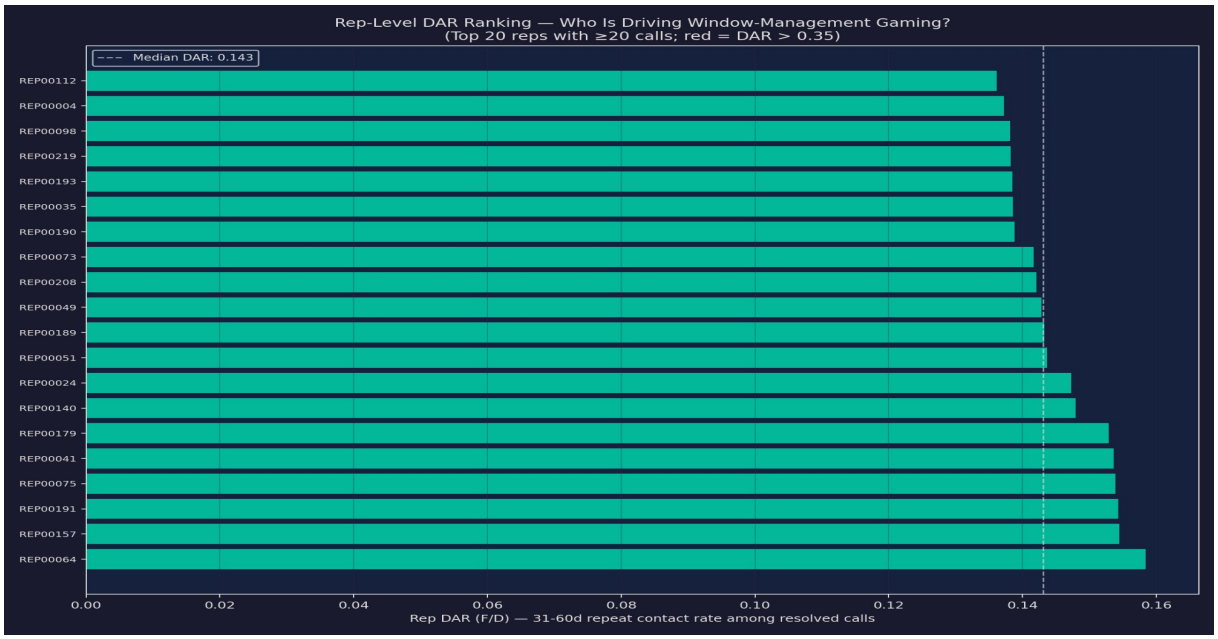

Figure 8. Rep-level DAR ranking (top 20 by highest DAR). All reps cluster near the median DAR of 0.143. No rep exceeds the 0.35 gaming threshold. The distribution indicates systemic call difficulty rather than individual gaming behavior.

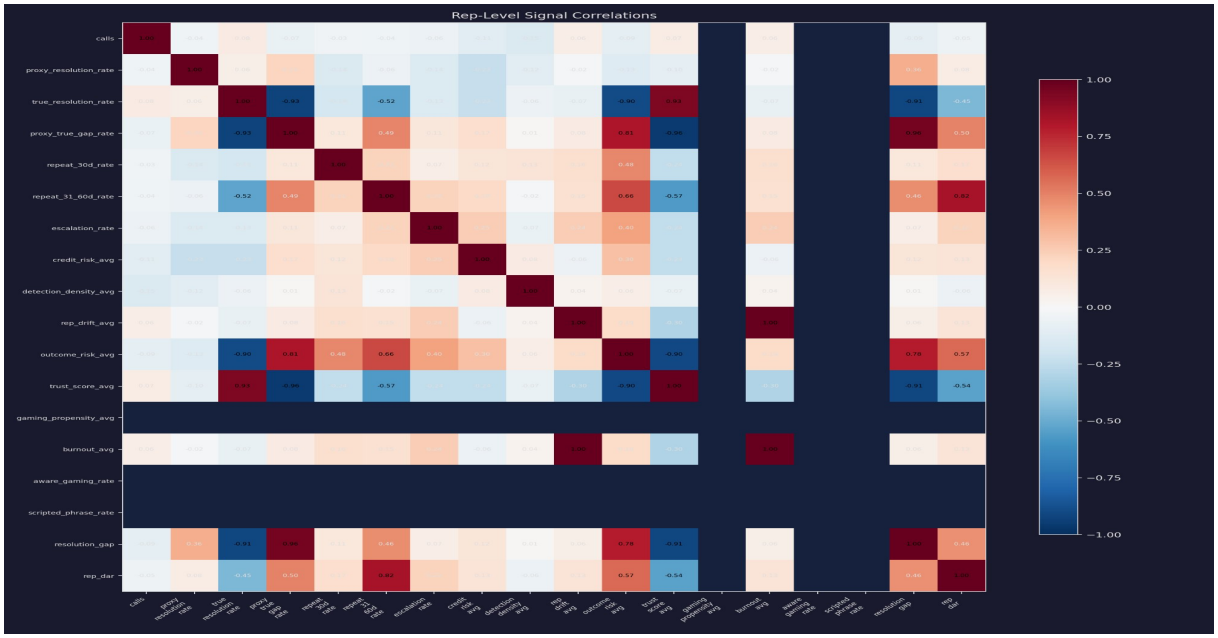

Figure 9. Rep-level signal correlation matrix. Gaming propensity and scripted phrase rate show near-zero correlations with all outcome variables, confirming absence of gaming patterns.

In the distorted environment, rep gaming propensity correlated strongly with the resolution gap ($r = +0.461$, $p < .001$). Here, gaming_propensity_avg and aware_gaming_rate show near-zero correlations with all outcome variables. The signal structure of a clean system looks fundamentally different from a gaming system.

9. Churn Integration

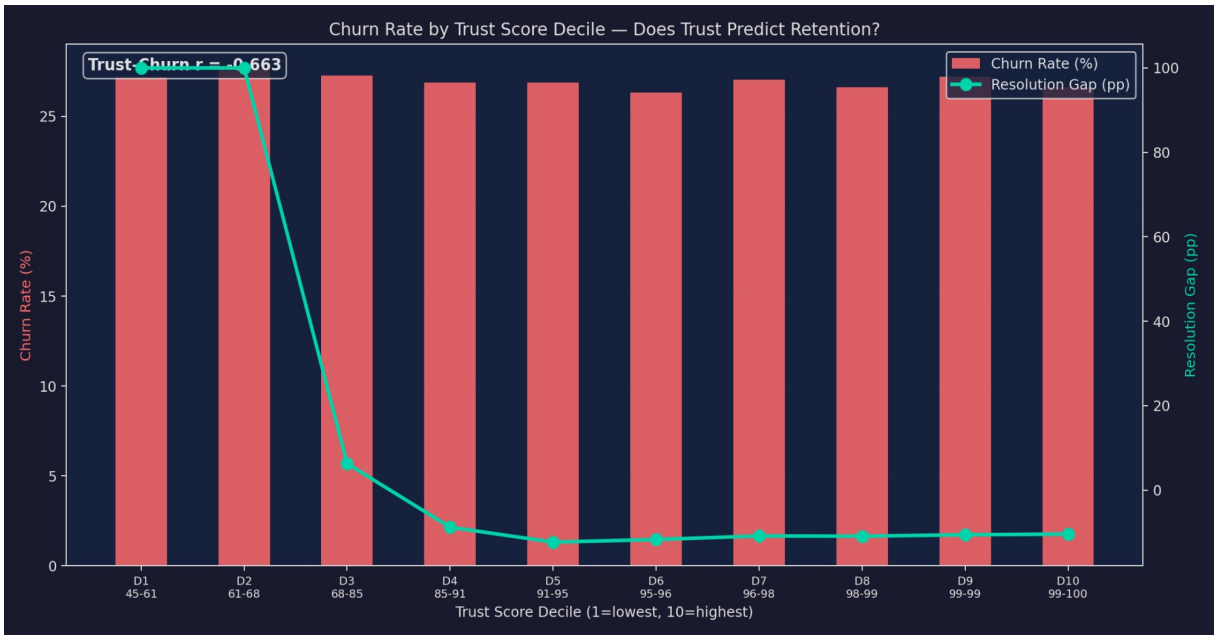

Figure 10. Churn rate by trust score decile. Trust-churn correlation r = -0.663. The resolution gap collapses after D2, confirming high-gap deciles are driven by structurally difficult calls, not gaming.

Trust score is predictive of churn at $r = -0.663$. D1 and D2 (trust scores 45-68) churn at 27.2% and 27.6% with a 100% resolution gap -- these are the unresolvable and activation-failed calls where no durable outcome is possible. From D3 onward, churn stabilizes and the resolution gap disappears or inverts. The system acknowledges difficulty at the bottom of the trust distribution rather than hiding it behind credits.

10. Credit Behavior

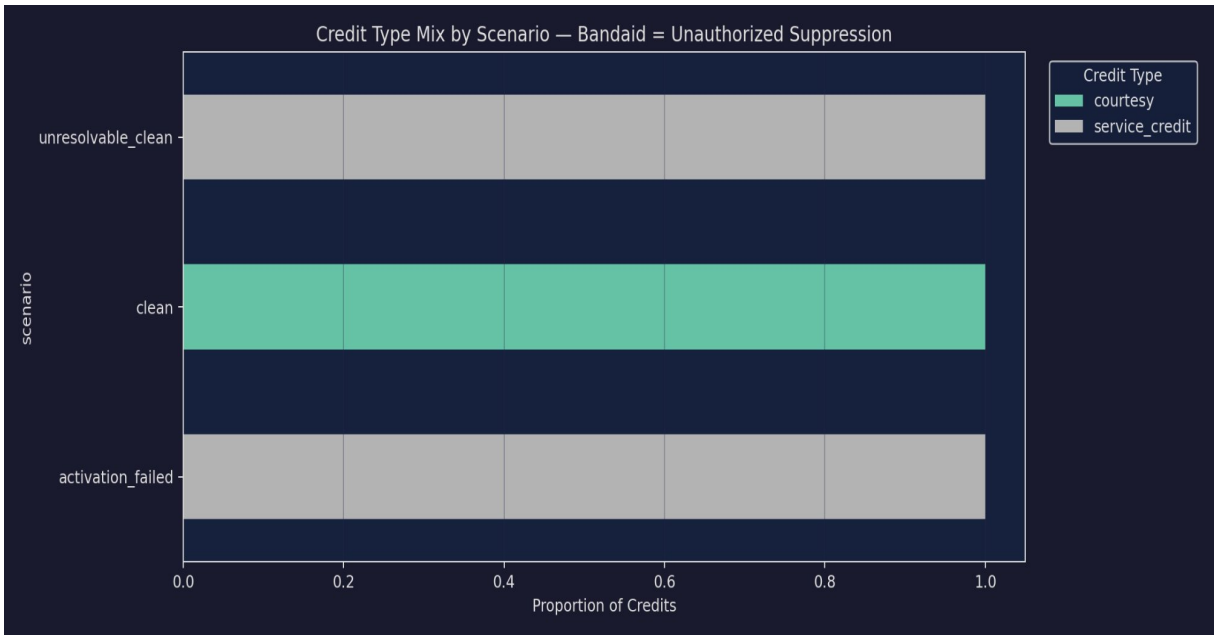

Figure 11. Credit type mix by scenario. Clean scenarios show courtesy credits only. Unresolvable and activation_failed show service credits -- appropriate compensation for genuine service failures. Zero bandaid credits across all scenarios.

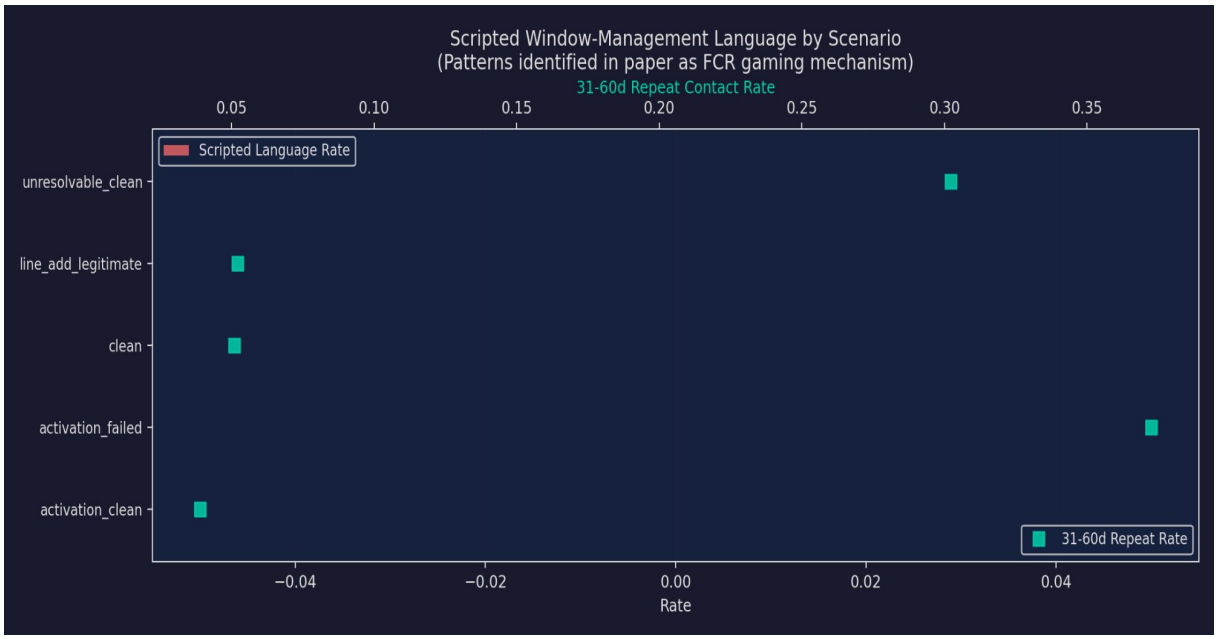

Figure 12. Scripted window-management language by scenario. Near-zero scripted language rates confirm absence of FCR gaming mechanisms.

The clean scenario shows only courtesy credits. Unresolvable_clean and activation_failed show service credits -- the correct response to a genuine service failure the rep cannot resolve. No scenario shows bandaid credits. The credit system is being used as designed.

11. Comparison with the Distorted Environment

Table 3 places the governance-aligned results alongside the key findings from the distorted environment (Aulabaugh, 2026a).

Metric	Distorted Environment	Governance-Aligned
Proxy FCR	89.4%	76.9%
True resolution	47.3%	63.8%
Resolution gap	42.1 pp	13.1 pp
Bandaid credit rate	71.1%	0.0%
SII	High (VETO range)	5.2 (OK)
VETO alerts	Fired	0
Trust score	Degrading over time	87.2 -- stable
TER vs. baseline	OR = 0.99, p = .78	delta = +0.0006
Gaming propensity x gap	r = +0.461, p < .001	~0 (no correlation)
Rep burnout	Increasing under pressure	Stable baseline

Table 3. Side-by-side comparison. Distorted environment data from Aulabaugh (2026a). Alternating row colors indicate distorted (pink) vs. governance-aligned (green) values.

The governance-aligned environment has a lower proxy FCR (76.9%) than the distorted environment (89.4%). This is counterintuitive until the mechanism is understood. In the distorted environment, gaming inflates the proxy by marking difficult calls as resolved when they are not. In the governance-aligned environment, those calls are handled honestly. A lower proxy FCR can be a sign of better governance.

12. Conclusions

This paper establishes five conclusions about the governance-aligned measurement environment:

1. **The Trust Signal Health framework correctly identifies a healthy system.**

   SII = 5.2, zero alerts, zero bandaid credits. The framework fires when distortion is present and is quiet when it is not.

2. **The 13.1 pp resolution gap is operationally explained, not a governance concern.**

   It is entirely attributable to unresolvable_clean and activation_failed scenarios. Clean scenarios show near-zero or negative gaps.

3. **Bandaid credit rate is the most sensitive governance discriminator.**

   0.0% here vs. 71.1% in the distorted environment. This single signal provides the clearest separation between aligned and misaligned systems.

4. **Trust scores are stable and churn-predictive.**

   r = -0.663 trust-churn correlation confirms the framework captures something real about outcome quality. Monthly stability confirms no accumulating drift.

5. **The governance baseline is now calibrated.**

   A SII of 5.2 under honest operations tells the analyst what the floor looks like. Anything substantially above this in a similar system warrants investigation.

A governance layer that can confirm alignment is more valuable than one that can only detect failure. Organizations deploying AI optimization in operational environments need both signals: the VETO that halts a runaway system and the baseline confirmation that the measurement environment remains trustworthy. This paper provides evidence that the Trust Signal Health framework delivers both.

References

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